



VJTech Academy
Inspiring Your Success...

“JAVA Practical Programs”



//Write a Java program to check whether number is EVEN or ODD

```
import java.util.*;
class CheckEvenOdd
{
    public static void main(String args[])
    {
        int no;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter any number:");
        no=sc.nextInt();
        if(no%2==0)
        {
            System.out.println("Number is EVEN");
        }
        else
        {
            System.out.println("Number is ODD");
        }
    }
}
```

//Write a Java program to make use of logical operator

```
import java.util.*;
class LogicalOperator
{
    public static void main(String args[])
    {
        boolean x;

        x=!((10>5&&10==20) || (6<7 || 9!=35));

        System.out.println("Result of !((10>5&&10==20) || (6<7 || 9!=35)) =>"+x);
    }
}
```

OUTPUT:

Result of !((10>5&&10==20) || (6<7 || 9!=35)) =>false

//Write a Java program to make use of switch case statement using character datatype.

//check whether alphabet is vowel or not.

```
import java.util.*;
class SwitchCase
{
    public static void main(String args[])
    {
        char ch;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter any alphabet:");
        ch=sc.next().charAt(0);
        switch(ch)
        {
            case 'a':
            case 'e':
            case 'i':
            case 'o':
            case 'u':
            case 'A':
            case 'E':
            case 'I':
            case 'O':
            case 'U':System.out.println("Alphabet is Vowel");
                    break;
            default: System.out.println("Alphabet is consonant");
        }
    }
}
```

OUTPUT:

Enter any alphabet:

a

Alphabet is Vowel

//Write any program using if condition with for loop

//WAP to display Even number between 1 to 50

```
class DisplayEvenNo
{
    public static void main(String args[])
    {
        int i;
        System.out.println("Even Number between 1 to 50:");
        for(i=1;i<=50;i++)
        {
            if(i%2==0)
            {
                System.out.println(i);
            }
        }
    }
}
```

/*WAP to display pyramids of stars/pattern using increment/decrement operator.

*

**

*/

```
class DisplayStar1
{
    public static void main(String args[])
    {
        int i,j;
        for(i=1;i<=5;i++)
        {
            for(j=1;j<=i;j++)
            {
                System.out.print("*");
            }
            System.out.println();
        }
    }
}
```

OUTPUT:

*

**

//WAP to display 1 to 50 numbers

```
class Display1to50Nos
{
    public static void main(String args[])
    {
        int i=1;
        do
        {
            System.out.println(i);
            i++;
        }while(i<=50);
    }
}
```

//WAP to implicitly typecast lower range datatype to larger storage size datatype

```
class ImplicitTypeCasting
{
    public static void main(String args[])
    {
        int x=100;
        float b;
        b=x;           //implicit
        System.out.println("Value of b="+b);
    }
}
```

//WAP to explicitly typecast lower range datatype to larger storage size datatype

```
class ExplicitTypeCasting
{
    public static void main(String args[])
    {
        int x=100;
        float b;
        b=(float)x;           //explicit type casting
        System.out.println("Value of b="+b);
    }
}
```

//WAP to implement different types of constructors to perform addition of complex no.

```
class Complex
{
    int real,imag;
    Complex()
    {

    }
    Complex(int r,int i)
    {
        real=r;
        imag=i;
    }
    void display()
    {
        System.out.println(real+"+i"+imag);
    }
    void Addition(Complex m,Complex n)
    {
        real=m.real+n.real;
        imag=m.imag+n.imag;
    }
    public static void main(String args[])
    {
        Complex c1=new Complex(100,200);
        Complex c2=new Complex(10,20);
        Complex c3=new Complex();
        c3.Addition(c1,c2);
        c1.display();
        c2.display();
        c3.display();
    }
}
```

OUTPUT

100+i200

10+i20

110+i220

//WAP to implement all methods of StringBuffer class

```
class StringBufferDemo
{
    public static void main(String args[])
    {
        StringBuffer str=new StringBuffer("VJTech");
        System.out.println("Original String :"+str);
        System.out.println("Length of String :"+str.length());
        System.out.println("Character at position 2 = "+str.charAt(2));
        str.setCharAt(3,'T');
        System.out.println("Modified String :"+str);
        str.append("Academy");
        System.out.println("Appended String :"+str);
        System.out.println("SubString :"+str.substring(3));
    }
}
```

OUTPUT

Original String :VJTech

Length of String :6

Character at position 2 = T

Modified String :VJTch

Appended String :VJTchAcademy

SubString :TchAcademy

//WAP to display array elements using for each loop

```
import java.util.*;

class ForEachLoopDemo
{
    public static void main(String args[])
    {
        int array[]=new int[5];
        int i;
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter five array elements:");
        for(i=0;i<5;i++)
        {
            array[i]=sc.nextInt();
        }
        System.out.println("Array Elements:");
        for(int p:array)
        {
            System.out.println(p);
        }
    }
}
```

//WAP to use different methods of vector class.

```
import java.util.*;
class VectorDemo
{
    public static void main(String args[])
    {
        Vector v1=new Vector();
        v1.addElement(new Integer(10));
        v1.addElement(new Integer(30));
        v1.addElement(new Integer(60));
        v1.addElement(new Integer(70));
        v1.addElement(new Integer(80));
        v1.addElement(new Integer(100));
        System.out.println("Original Size of Vector:"+v1.size());
        v1.removeElementAt(3);
        v1.removeElementAt(4);
        v1.insertElementAt(new Integer(50),3);
        System.out.println("Current Size of Vector:"+v1.size());
        System.out.println("Vector Elements:"+v1);
    }
}
```

OUTPUT

```
Original Size of Vector:6
Current Size of Vector:5
Vector Elements:[10, 30, 60, 50, 80]
```

/*Method overriding: WAP to create Dog class which extends Animal class to override move method using super keyword*/

```
class Animal
{
    void move()
    {
        System.out.println("move method of Animal class");
    }
}
class Dog extends Animal
{
    void move()
    {
        super.move(); //call method override
        System.out.println("move method of Dog class");
    }
    public static void main(String args[])
    {
        Dog d1=new Dog();
        d1.move();
    }
}
```

OUTPUT

```
move method of Animal class
move method of Dog class
```

/*Develop a program to calculate the room area and volume using single inheritance.

class

area=Length*Width

volume=(Length*Width*Height)

*/

```
import java.util.*;

class Base
{
    int Length,Width,Height;
    void getdata()
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Room Length:");
        Length=sc.nextInt();
        System.out.println("Enter Room Width:");
        Width=sc.nextInt();
        System.out.println("Enter Room Height:");
        Height=sc.nextInt();
    }
}

class Derived extends Base
{
    void display()
    {
        System.out.println("Area of Room="+Length*Width);
        System.out.println("Volume of Room="+Length*Width*Height);
    }
    public static void main(String args[])
    {
        Derived d1=new Derived();
    }
}
```

```
        d1.getdata();  
        d1.display();  
    }  
}
```

OUTPUT

```
Enter Room Length:  
10  
Enter Room Width:  
20  
Enter Room Height:  
30  
Area of Room=200  
Volume of Room=6000
```

//MultilevelInheritanceDemo student->Test->Result class

```
import java.util.*;

class Student
{
    int rollNo;
    String name;
    void get_stud_info()
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Student Roll No:");
        rollNo=sc.nextInt();
        System.out.println("Enter Student Name:");
        name=sc.next();
    }
    void disp_stud_info()
    {
        System.out.println("Student ROLL No:"+rollNo);
        System.out.println("Student Name:"+name);
    }
}

class Test extends Student
{
    int marks1,marks2;
    void get_stud_marks()
    {
        Scanner sc=new Scanner(System.in);
        System.out.println("Enter Student Test-1 Marks:");
        marks1=sc.nextInt();
        System.out.println("Enter Student Test-2 Marks:");
        marks2=sc.nextInt();
    }
}
```

```
        }

        void disp_stud_marks()
        {
            System.out.println("Test-1 Marks:"+marks1);
            System.out.println("Test-2 Marks:"+marks2);
        }
    }

class Result extends Test
{
    int total;
    void display_total_marks()
    {
        total=marks1+maks2;
        System.out.println("Total Marks:"+total);
    }
}

class MultilevelInheritanceDemo1
{
    public static void main(String args[])
    {
        Result r1=new Result();
        r1.get_stud_info();
        r1.get_stud_marks();
        System.out.println("*****STUDENT INFORMATION SYSTEM*****");
        r1.disp_stud_info();
        r1.disp_stud_marks();
        r1.display_total_marks();
    }
}
```


OUTPUT:

```
Enter Student Roll No:
1010
Enter Student Name:
James
Enter Student Test-1 Marks:
99
Enter Student Test-2 Marks:
67
*****STUDENT INFORMATION SYSTEM*****
Student ROLL No:1010
Student Name:James
Test-1 Marks:99
Test-2 Marks:67
Total Marks:166
```

//Multiple Inheritance

```
interface A
{
    void displayA();
}
interface B
{
    void displayB();
}
class C implements A,B
{
    public void displayA()
    {
        System.out.println("displayA method of A class");
    }
    public void displayB()
    {
        System.out.println("displayB method of B class");
    }
    public static void main(String args[])
    {
        C c1=new C();
        c1.displayA();
        c1.displayB();
    }
}
```

OUTPUT

```
displayA method of A class
displayB method of B class
```

/*Develop a package named myInstitute include class named as department with one method to display the staff of that department. Develop program to import this package in a java application and call the method defined in the package*/

```
package myInstitute;

public class department
{
    public void display()
    {
        System.out.println("***STAFF INFORMATION***");
        System.out.println("Staff Name: James Gosling");
        System.out.println("Staff Code: 1010");
        System.out.println("Staff Post: Teacher");
        System.out.println("Staff Salary: 35000");
    }
}
```

```
import myInstitute.*;
class AccessmyInstitutePKG
{
    public static void main(String args[])
    {
        department d1=new department();
        d1.display();
    }
}
```

OUTPUT

```
***STAFF INFORMATION***
Staff Name: James Gosling
Staff Code: 1010
Staff Post: Teacher
Staff Salary: 35000
```